QSAR CHALLENGE: PREDICTING MOLECULAR ACTIVITY FROM SMILES

LUKAS STEU

UE ARTIFICIAL INTELLIGENCE IN LIFE SCIENCES

GOAL: PREDICT BIOLOGICAL ACTIVITY FOR 11 TASKS FROM MOLECULAR STRUCTUR

THE PROBLEM

What is the problem?	• Predict biological activity or toxicity from a molecule's structure
Who has this problem?	• Me, pharma and biotech companies
Why should this problem be solved?	• Can accelerate drug discovery and reduce unnecessary experiments
How will I know this problem has been solved?	• When the model performs well on unseen test molecules

BACKGROUND INFORMATION 1

- QSAR = Quantative Structure-Activity Relationship
  - Input: molecule as SMILES String
  - Output: activity prediction for 11 tasks
  - Label meaning
    - +1 = active
    - -1 = inactive
    - 0 = unknown
    - Importance: unknown labels are NOT negative labels and must be ignored per task

DATASET AND CHALLENGE SETUP

- Training data:
  - SMILES + 11 Task Labels
- Test data:
  - SMILES only
- Submission:
  - One score per task and molecule
- Evaluation:
  - Mean ROC-AUC across tasks
  - Unknown labels masked out

A1 | fx | ,smiles,task1,task2,task3,task4,task5,task6,task7,task8,task9,task10,task11

	A
1	,smiles,task1,task2,task3,task4,task5,task6,task7,task8,task9,task10,task11
2	0,O=C(O)COc1nn(Cc2ccccc2)c2ccccc12,-1,0,0,-1,0,0,0,-1,-1,1,-1
3	1,COc1cc(F)c(C(C)C)cc1-c1ccc(C(F)(F)F)cc1CN1C(=O)O[C@H](c2cc(C(F)(F)F)cc(C(F)(F)F)c2)[C@H]1C,0,0,0,0,0,0,0,0,0,0,-1
4	2,CCCCCCCCCCCCCCCCC(=O)OCC(CC)CCCC,-1,0,0,0,0,0,0,-1,0,0,0

A1 | fx | ,smiles

	A
1	,smiles
2	0,Cc1cc(-c2nncc3c(C(C)(C)C)c(OCc4ncnn4C)nn23)no1
3	1,C=CCCCCCCCCC(=O)OCC
4	2,[2H]C([2H])([2H])C(O)(CCO[C@@H](C)[C@H]1CC[C@H]2/C(=C/C=C3/C[C@@H](O)C[C@H](O)C3=C)CCC[C@]12C)C([2H])([2H])[2H]

DATA PREPROCESSING AND REPRESENTATION

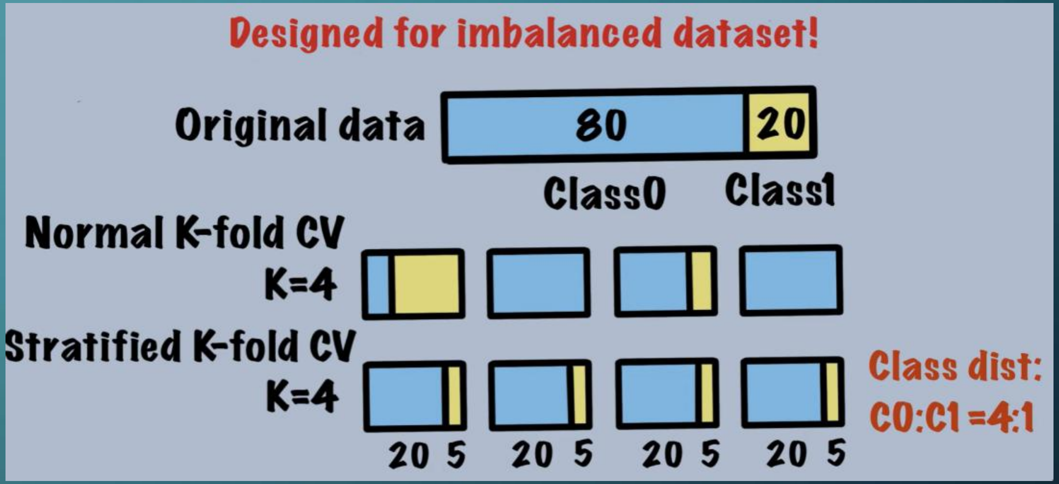
- SMILES parsed with RDKit
- Molecules represented by:
  - Morgan fingerprints
    - Caputre local substructures, widely used QSAR baseline, robust and fast
  - Optional RDKit descriptors
    - Gobal chemical properties: molecular weight, logP, TPSA, H-bond donors/acceptors etc.
- Invalid molecules were not removed
- Fallback strategy used to preserve all rows

MODELING STRATEGY

- Treated as 11 separate binary classification tasks
- For each task:
  - Keep only known labels
  - Map +1 to 1 and -1 to 0
  - Train one classifier
- Baseline models tested:
  - Logistic Regression, ExtraTrees and Random Forest
  - Why: simple, robust, easy handling of missing labels and (at least for me) easy to use ☺
- Prediction:
  - Use probability scores, not hard class labels

VALIDATION STRATEGY

- Used Stratified K-Fold Cross-Validation
  - Imbalanced dataset
- Generated out-of-fold predictions (OOF)
- Computed AUC per task
- Final score = mean AUC across tasks



Stratified vs Normal K-Fold

Source: https://www.youtube.com/watch?v=mMKyIjcoaCE

RESULTS

Logistic Regression

- Features:
  - Morgan + descriptors
  - Radius: 2 & bits: 2048
- Mean OOF AUC
  - 0.75622
- Runtime
  - 27s

ExtraTrees

- Features:
  - Morgan + descriptors
  - Radius: 2 & bits: 2048
- Mean OOF AUC
  - 0.76250
- Runtime:
  - 3m 10s

Random Forests

- Features:
  - Morgan + descriptors
  - Radius: 2 & bits: 2048
- Mean OOF AUC
  - 0.78053
- Runtime:
  - 1m 53s

KEY FINDINGS & INTERPRETATION

- Morgan fingerprints provide a strong baseline for QSAR
- handling missing labels correctly is crucial
- simple models can already achieve competitive AUC
- some tasks are easier than others, likely due to data quantity or separability
- descriptors help, but their effect should be validated empirically

LIMITATIONS & FUTURE WORK

- Limitations:
  - Random CV may overestimate generalization in QSAR
  - Class imbalance can affect training stability
  - Fixed fingerprints may lose some structural nuance
  - Separate task models do not exploit task correlations
- Future Work:
  - Scaffold based validation
  - Hyperparameter tuning
  - Model ensembling
  - Gradient boosting methods
  - Graph neural networks / message passing networks